Educational Pathways to Health: Early Life Family Environment as a Moderator Across Racial/Ethnic Groups

Abstract

This study examines how early life family environments moderate the relationship between educational attainment and cardiometabolic risk (CMR) across racial/ethnic groups in the United States. Using data from the National Longitudinal Study of Adolescent to Adult Health (Add Health), we focus on several key family factors: family cohesion, parental expectations for education, parenting style, parental education, parental income, and family structure. Our findings reveal that the association between educational attainment and CMR varies substantially by race/ethnicity and is significantly moderated by family environment. White respondents demonstrate robust health benefits from higher education regardless of family context, while Black respondents show significant health improvements from higher education only in specific family environments characterized by high cohesion, high educational expectations, and two-parent families. Counterintuitively, Black respondents from families with autonomic parenting styles, lower parental education, and lower parental income experience greater health benefits from higher education than those from more socioeconomically advantaged backgrounds. For US-Born Hispanics, education's health benefits are most pronounced in contexts of high parental income and expectations, while Foreign-Born Hispanics show more limited patterns of educational health benefits. These findings underscore the importance of family context in shaping health trajectories and highlight the need for targeted interventions to reduce racial health disparities that consider both educational attainment and family background.

Keywords: Education, Cardiometabolic Risk, Family Environment, Racial Disparities, Socioeconomic Status

Introduction

Educational attainment significantly influences health outcomes. Higher education is generally linked to better health and longevity (Conti et al., 2010; Montez et al., 2012). However, this relationship varies across racial/ethnic and nativity groups (Assari et al., 2016; Beltrán-Sánchez et al., 2016; Boen, 2016). For instance, highly educated Black Americans experience fewer health benefits compared to their White counterparts, a phenomenon called "diminishing returns" (Farmer et al., 2020). In contrast, some Hispanic subgroups exhibit comparable or better health outcomes despite lower average socioeconomic status (Markides & Rote, 2015), often termed as the "Hispanic paradox."

Recent research has emphasized the role of contextual factors in shaping the education-health relationship. Montez et al. (2012) revealed state-level variations in educational health disparities, while Hargrove et al. (2022) demonstrated how county-level characteristics moderate the association between education and cardiometabolic risk. These studies highlight the impact of broader socioeconomic and policy contexts on the benefits of educational attainment for adulthood health.

The theory of Fundamental Social Causes (FCT) provides a framework for understanding these contextual influences. The education-health association is not inherently causal but profoundly shaped by social and epidemiological contexts. Education likely has a causal effect on health primarily in contexts that stratify health-related risks and resources by education level (Montez & Bisesti, 2024). Individuals are embedded in historical, socioeconomic, and policy environments that influence the extent to which education matters for health (Montez et al., 2012).

While research has begun to explore the role of geographical contexts in educational health disparities, there remains a significant gap in understanding how family contexts might moderate the relationship between educational attainment and adult health outcomes across different racial/ethnic groups. Family environments represent critical microsystems where individuals develop capabilities and behaviors that shape health trajectories (Friedman et al., 2015). However, systematic differences in family environments across racial/ethnic groups may contribute to the inconsistent education-health relationship observed among these populations.

This study addresses this gap by examining how early life family environments moderate the relationship between educational attainment and cardiometabolic risk in young adulthood across different racial/ethnic and nativity groups. Using data from the National Longitudinal Study of Adolescent to Adult Health (Add Health), we address three key research questions:

1. How does the association between educational attainment and cardiometabolic risk vary across different race/ethnicity groups?

2. To what extent do family-level factors - including socioeconomic status, family structure, and psychosocial resources - explain variations in the education-health relationship?
3. Does the moderating role of family environment differ across racial/ethnic and nativity groups?

We focus on cardiometabolic risk as our health outcome because it represents a crucial pathway through which social factors influence long-term health trajectories. Cardiometabolic risk factors emerge early in life and are sensitive to both socioeconomic conditions and psychosocial environments. Moreover, racial/ethnic disparities in cardiometabolic health are well-documented and contribute substantially to overall health inequalities.

Background

2.1 Education and Health Outcomes Across Racial/Ethnic-Nativity Groups

Educational attainment has long been recognized as one of the strongest and most consistent predictors of health outcomes. A substantial body of research demonstrates that individuals with higher levels of education experience healthier and longer lives compared to those with lower levels of education (Hayward et al., 2015; Zajacova & Lawrence, 2018).

However, this relationship varies significantly across racial/ethnic and nativity groups. The association between education and health tends to be weaker or nonsignificant for Black adults relative to White adults across various health outcomes, including self-rated health, disability, and inflammation (Assari et al., 2016; Boen, 2016; Fuller-Rowell et al., 2015). This pattern, "diminishing returns," suggests that highly educated Black Americans receive fewer health benefits from educational attainment compared to White Americans.

Additionally, Hispanic and immigrant groups often exhibit more favorable health outcomes than their White and native-born counterparts despite lower average educational attainment (Goldman et al., 2006; Markides & Rote, 2015), a phenomenon known as the "Hispanic paradox." These inconsistencies challenge simplistic understandings of education as a health determinant and suggest that contextual factors may moderate how education translates into health advantages.

These findings underscore the complex nature of educational disparities in health and raise important questions about mechanisms that operate differently across racial/ethnic and nativity groups. While differences in the education-health relationship are well-documented, explanations for these differences remain incomplete, pointing to the need for a framework that considers both individual and contextual influences on health outcomes.

2.2 Proposed Explanations for Educational Disparities in Health

Several explanations have been proposed for the inconsistent relationship between education and health across racial/ethnic and nativity groups. For Black Americans, the "diminishing returns" phenomenon has been attributed to unique stressors encountered in the pursuit of higher education. Highly educated Black individuals often experience increased exposure to racial discrimination in predominantly White educational and occupational settings (Hudson et al., 2012; Miller et al., 2016). The psychological and physiological toll of navigating these environments may undermine the health benefits typically associated with educational attainment (Geronimus et al., 2016; Hudson et al., 2016; James et al., 1983). Additionally, the increased effort required to achieve educational success in the face of structural barriers may generate physical health costs for Black Americans (Brody et al., 2013; Colen et al., 2018).

For Hispanic populations, the "Hispanic paradox" suggests that cultural factors such as stronger family ties, healthier dietary practices, and lower rates of risky behaviors may protect health despite lower socioeconomic status (Markides & Rote, 2015). Among foreign-born Hispanics, selective migration patterns may also play a role, as those who migrate tend to be healthier than those who remain in their countries of origin (Kennedy et al., 2015; Riosmena et al., 2013). These sociocultural and migration-related factors may explain why foreign-born Hispanics with low education levels often exhibit better health than their similarly educated counterparts from other racial/ethnic groups.

While these explanations provide valuable insights, they have primarily focused on individual-level mechanisms, with less attention to intermediate contexts and broad societal factors. The individual-level focus assumes that educational disparities in health arise primarily from the characteristics, behaviors, and resources of individuals, rather than from the contexts in which education is obtained and translated into health advantages.

The Theory of Fundamental Social Causes (FCT) offers a broader framework that can help bridge individual and contextual explanations (Link & Phelan, 1995; Phelan & Link, 2015). FCT posits that social disparities in health exist when resources that can prevent disease and premature death (e.g., money, knowledge, power, social connections) are unevenly distributed in the population. However, recent extensions of FCT emphasize that the education-health relationship is not inherently causal but profoundly shaped by social and epidemiological contexts (Hayward et al., 2015; Montez & Friedman, 2015).

As Hayward et al. (2015) assert, education likely has a causal effect on health primarily in contexts that stratify health-related risks and resources by education level. This contextual perspective recognizes that individuals are embedded within historical, socioeconomic, and policy environments that determine the extent to which education matters for health. The mechanisms linking education to health are constantly evolving and context-dependent; as one mechanism is eliminated, another emerges to take its place (Montez & Bisesti, 2024). This helps explain why educational disparities in health persist and often grow over time, despite changes in specific health risks and resources.

Recent research has begun to explore the role of intermediate contexts in shaping educational health disparities. Studies have demonstrated that the relationship between education and health varies across geographic contexts such as states (Montez, Hayward, et al., 2019; Montez, Zajacova, et al., 2019) and counties (Hargrove et al., 2022). Gaydosh and Harris (2024) also demonstrate that educational institutions function as racialized organizations where college health benefits are consistently realized for White graduates across institution types, while Black graduates from elite institutions experience higher cardiometabolic risk comparable to non-college graduates. These studies show that state policy environments, county-level economic opportunities and school characteristics can significantly moderate the health benefits of education. For instance, the education-health association is stronger in states with less generous social safety nets and in counties with more limited economic mobility.

Despite this growing attention to geographic contexts, the role of family environments in moderating the education-health relationship remains understudied. Family represents a primary microsystem where individuals develop capabilities, behaviors, and resources that influence both educational attainment and health trajectories. Families shape access to resources, exposure to stressors, and development of health-related behaviors from early life through adulthood (Friedman et al., 2015; Montez & Hayward, 2014). However, the role of family environments in moderating the education-health relationship across racial/ethnic groups represents a significant gap in our understanding of educational health disparities.

2.3 The Role of Family Environment in Education-Health Relationships

Family environments influence health through multiple pathways. Family socioeconomic status affects access to educational and health resources (Chetty et al., 2020). Family structure impacts stress levels and support systems available for educational pursuits (McLanahan & Percheski, 2008). Parenting practices shape educational aspirations and health behaviors (Darling & Steinberg, 1993). Additionally, families provide varying levels of psychosocial resources that may moderate the education-health relationship (Hardie & Seltzer, 2016).

The family context may be particularly important for understanding racial/ethnic disparities in the education-health relationship. Structural racism leads to unequal distribution of resources within families, creating additional stressors for minoritized populations (Williams et al., 2016). These structural inequalities may constrain families' ability to convert higher education into health benefits for their children.

Limited indirect evidence suggests possible mechanisms through which family environments might moderate the education-health relationship across racial/ethnic groups. Chetty et al. (2020) found disparities in economic mobility across racial groups even in similar neighborhoods. Rothon et al. (2012) showed that family social capital differentially affects health outcomes across populations. Assari (2018; 2016) demonstrated that parental education is more strongly associated with lower depressive symptoms for White than Black youth.

These studies suggest that family background may shape how education translates to health differently across racial/ethnic groups. However, a comprehensive examination of how various dimensions of family environment moderate the education-health relationship across diverse populations is lacking. Addressing this gap could provide crucial insights into persistent racial/ethnic health disparities and inform more targeted interventions to reduce health inequalities.

2.4 The Present Study

This study uses longitudinal data to examine whether early life family environment moderates the relationship between educational attainment and cardiometabolic risk in young adulthood across different racial/ethnic and nativity groups. The theories and research outlined thus far lead to the following hypotheses corresponding with our three research questions:

> *Hypothesis 1*: While higher levels of education will be associated with better health for all racial/ethnic-nativity groups, the education-health relationship will be stronger among White and foreign-born Hispanic young adults than among Black and U.S.-born Hispanic young adults.
>
> *Hypothesis 2*: The education-health relationship will be most comparable across racial/ethnic-nativity groups in families characterized by economic, structural, and psychosocial advantages.
>
> *Hypothesis 3*: Across all early life family environments, higher educational attainment will be associated with better health among White young adults. Higher educational attainment will be associated with better health for Black and Hispanic young adults relative to their less educated counterparts of the same race/ethnicity and nativity status only when they are exposed to advantaged family environments in early life.

This research contributes to our understanding of educational health disparities in several ways. First, it moves beyond examining simple education-health associations to investigate how family contexts modify these relationships. Second, it considers multiple dimensions of family environment, providing a more comprehensive understanding of how families influence health trajectories. Third, by examining these patterns across racial/ethnic and nativity groups, it helps identify potential mechanisms underlying persistent health disparities. Finally, this work has important implications for interventions aimed at reducing health inequalities, suggesting that efforts to promote health through education must consider the crucial role of family contexts.

Data and Methods

3.1 Data Source and Sample

This study uses data from Waves I and IV of the National Longitudinal Study of Adolescent to Adult Health (Add Health). Wave I data were collected in 1994-1995 from a nationally representative sample of adolescents in grades 7-12, including both in-school questionnaires

and in-home interviews with adolescents and their parents. Wave IV follow-up data were collected in 2008-2009 when respondents were aged 24-32, incorporating extensive biomarker collection and health assessments.

The initial Wave I sample included 20,745 adolescents. At Wave IV, 15,701 original respondents were re-interviewed. Our analytic sample consists of 9,129 respondents who:

1. Participated in both Wave I and Wave IV

2. Have valid sampling weights

3. Have complete biomarker data for constructing the cardiometabolic risk index and for predictors of interest

4. Identify as non-Hispanic White, non-Hispanic Black, or Hispanic

3.2 Measures

Outcome Variable

The primary outcome measure is a cardiometabolic risk (CMR) index constructed from seven biomarkers collected during Wave IV medical examinations. These biomarkers include waist circumference, blood pressure, lipid profiles (triglycerides, HDL cholesterol, and LDL cholesterol), glycated hemoglobin (HbA1c), and C-reactive protein (CRP). Following established clinical guidelines, each biomarker was dichotomized to indicate elevated risk. Waist circumference was assessed through standardized measurements and considered elevated at ≥102 cm for men and ≥88 cm for women. Blood pressure measurements were taken while participants were seated, with elevated risk defined as systolic pressure ≥130 mmHg, diastolic pressure ≥85 mmHg, or current use of antihypertensive medication (based on the question "Are you currently taking any medication for high blood pressure?"). For lipids, elevated risk was defined as triglycerides ≥150 mg/dL, HDL cholesterol <40 mg/dL for men and <50 mg/dL for women, or LDL cholesterol ≥130 mg/dL. HbA1c values ≥5.7% or current use of diabetes medication indicated elevated risk, and CRP was considered elevated at >3 mg/L. These binary indicators were summed to create a count index ranging from 0 to 7, with higher scores indicating greater cardiometabolic risk.

Predictors of Interest

Our predictors of interest include race/ethnicity, nativity, educational attainment, and family environment characteristics measured during adolescence.

Self-identified race/ethnicity is measured by respondents' choice of racial/ethnic category: non-Hispanic White, non-Hispanic Black, and Hispanic. Nativity is assessed as whether the respondent was born outside of the United States. Given that few Black and White respondents were born outside of the United States in the Add Health cohort, we assess differences by nativity only among Hispanics. Thus, four racial/ethnic-nativity groups are considered: non-Hispanic U.S.-born Blacks (hereafter referred to as Black), U.S.-born

Hispanics, foreign-born Hispanics, and non-Hispanic U.S.-born Whites (hereafter referred to as White).

Educational attainment is measured as a dummy variable: 0 = less than a four-year college education and 1 = a four-year college education or more.

Family environment is captured through multiple measures from Wave I:

Parental Education is measured using the highest level of education achieved by either parent, dichotomized to indicate whether at least one parent completed college (1) or neither parent completed college (0). This coding parallels the educational attainment measure for respondents to facilitate direct comparison of intergenerational educational patterns.

Parental Income is measured using the parent questionnaire item: "About how much total income, before taxes did your family receive in 1994? Include your own income, the income of everyone else in your household, and income from welfare benefits, dividends, and all other sources." Responses ranged from $0 to $999 thousand. We examine parental income separately from parental education to differentiate the effects of different socioeconomic status components.

Parents' Expectations for Education is measured using the item: "On a scale of 1 to 5, where 1 is low and 5 is high, how disappointed would she/he be if you did not graduate from college?" We average the responses for both parents when available.

Parenting Style is captured through a series of items assessing parental autonomy granting: "Do your parents let you make your own decisions about: the time you must be home on weekend nights; the people you hang around with; what you wear; how much television you watch; which television programs you watch; what time you go to bed on week nights; what you eat?" We average responses across these items, with higher scores indicating greater autonomy granting. This measure draws on Lareau's (2011) insights into class differences in parenting approaches, where middle-class parents tend to engage in "concerted cultivation" while working-class parents follow a "natural growth" approach.

For Parenting style, income and expectations for education, we dichotmize the measures at race/ethnicity-specific 75th percentiles to identify the advantaged vs disadvantaged contexts.

Family Conflict is measured using questions about the frequency of serious arguments between respondents and their parents. Responses ranged from "never" to "almost every day" and were averaged across parents where applicable, then dichotomized at the median to indicate high conflict (0) versus low conflict (1).

Two-Parent Family is constructed from the household roster data as a binary indicator. Respondents living with two biological or adoptive parents were coded as 1, and all other arrangements (including single-parent households, stepfamilies, and other family configurations) were coded as 0.

Control Variables

Our models include both individual-level and contextual controls measured at Wave I. Individual controls include age (continuous, derived from date of birth), gender (female=1), and self-rated health (SRH, measured on a 5-point scale from "poor" (1) to "excellent" (5) in response to the question "In general, how is your health?"). County-level contextual factors include median household income (in thousands of dollars, obtained from Census data), educational composition (percentage of residents aged 25 and older with a college degree), and crime rate (total crimes per 100,000 residents, obtained from FBI Uniform Crime Reports). These contextual controls account for broader socioeconomic and environmental conditions that might influence both educational attainment and health outcomes.

3.3 Analytical Strategy

Our analysis employs a multi-stage approach to examine how early life family environment moderates the relationship between educational attainment and cardiometabolic risk across different racial/ethnic and nativity groups. We begin by conducting preliminary analyses to determine the appropriate model specification for our count outcome (CMR index). Tests of equidispersion reveal that the variance of our CMR measure is less than or equal to its mean, satisfying the key assumption of Poisson regression. This finding, combined with the discrete, non-negative nature of our outcome variable, supports our choice of survey-weighted Poisson regression as the primary analytical method.

The analysis proceeds In three stages. First, we estimate baseline models examining the fundamental association between educational attainment and cardiometabolic risk, controlling for individual and county characteristics, and we include interaction terms between educational attainment and race/ethnicity-nativity to assess how the education-health relationship varies across groups.

$$\ln\left[E(CMR_i|X_i)\right] = \beta_0 + \beta_1 Education_i + \beta_2 Race_i + \beta_3 Family_i + \beta_4 (Education \times Race)_i + \beta_8 Controls_i + \varepsilon_i \text{ (Model 1)}$$

Second, we systematically examine the moderating role of family environment through a series of models incorporating three-way interactions between education, race/ethnicity-nativity, and each family environment measure. These models take the following form:

$$\ln\left[E(CMR_i|X_i)\right] = \beta_0 + \beta_1 Education_i + \beta_2 Race_i + \beta_3 Family_i + \beta_4 (Education \times Race)_i + \beta_5 (Education \times Family)_i + \beta_6 (Family \times Race)_i + \beta_7 (Education \times Race \times Family)_i + \beta_8 Controls_i + \varepsilon_i \text{ (Model 2)}$$

Where CMR_i represents the cardiometabolic risk index for individual *I*, $Education_i$ indicates college degree attainment, $Race_i$ is a vector of race/ethnicity-nativity indicators, $Family_i$ represents each family environment measure, and $Controls_i$ includes both individual and

county-level covariates. The coefficient β_7 captures our key parameter of interest: how the moderating effect of family environment on the education-health relationship varies across racial/ethnic groups.

Additionally, we stratify our models by race/ethnicity-nativity and family environment characteristics to examine educational disparities in health within specific subgroups and contexts. This approach allows us to identify the conditions under which education has the strongest association with health for each racial/ethnic-nativity group.

$$\ln\left[E(CMR_i|X_i)\right] = \beta_0 + \beta_1 Education_i + + \beta_8 Controls_i + \varepsilon_i \text{ (Model 3)}$$

To facilitate interpretation of these complex interactions, we compute and plot predicted CMR scores across different combinations of education, race/ethnicity-nativity, and family environment characteristics, holding other variables at their means. We also calculate average marginal effects to quantify the magnitude of educational differences in CMR across different family contexts and racial/ethnic groups. Throughout our analysis, we incorporate Wave IV cross-sectional weights to account for unequal probability of selection and differential attrition, and we adjust standard errors for clustering and stratification in the Add Health sampling design. All analyses are conducted using Stata 18.0, with significance assessed at the conventional $\alpha = 0.05$ level using two-tailed tests.

Results

4.1 Sample Statistics

Table 1. Weighted means and proportions by race, ethnicity and nativity

	NH White (*n* = 5,991)	NH Black (*n* = 1,911)	US-Born Hispanic (*n* = 982)	Foreign-Born Hispanic (*n* = 245)
CMR	1.89	2.2	2.26	1.91
	(0.02)	(0.10)	(0.06)	(0.07)
Female	0.48	0.52	0.5	0.52
W1 SRH	2.07	2.06	2.06	2.22
	(0.03)	(0.04)	(0.07)	(0.05)
W1 Age	15.4	15.37	16.17	15.56
	(0.10)	(0.12)	(0.22)	(0.16)
Family Cohesion	0.67	0.69	0.77	0.68
Parent's Expectations on Study	3.28	3.67	3.38	3.21
	(0.02)	(0.03)	(0.10)	(0.04)
Parenting Style	5.34	5.06	4.65	4.9
	(0.05)	(0.07)	(0.22)	(0.07)
Parent College	0.4	0.32	0.21	0.17
Parental Income	54219.75	33144.09	20794.99	34011.72
	(1820.45)	(1432.33)	(1713.10)	(1469.36)
Two Parent Family	0.6	0.35	0.56	0.59
W1 County Crime Rate	5273.07	7278.25	8969.41	7198.98
	(203.91)	(203.73)	(915.82)	(482.58)
W1 County Median Income	31126.24	28104.72	31014.5	32421.05
	(641.56)	(774.33)	(1286.66)	(648.68)
W1 County Aged 25+ with College +	0.27	0.25	0.29	0.3
	(0.01)	(0.01)	(0.01)	(0.01)

Table 1 presents weighted descriptive statistics for the analytic sample by race/ethnicity-nativity group. For our primary outcome, cardiometabolic risk (CMR), notable disparities exist across groups, with Non-Hispanic White and Foreign-Born Hispanic respondents showing lower (more favorable) CMR scores compared to Non-Hispanic Black and US-Born Hispanic respondents. These patterns align with the "Hispanic paradox" where foreign-born Hispanics often display better health outcomes than their U.S.-born counterparts. Family environment measures also reveal substantial differences across groups. White respondents report the highest parental income and parental college completion rates, while Hispanic respondents report the lowest parental income and education levels. Black respondents are least likely to have grown up in two-parent families compared to other groups. Black respondents report the highest levels of parents' expectations for education, while White respondents report the highest levels of parental autonomy granting. These socioeconomic

and family structure differences may contribute to disparities in how education translates to health outcomes across racial/ethnic-nativity groups.

4.2 Racial/Ethnic-Nativity Differences in the Education–CMR Relationship

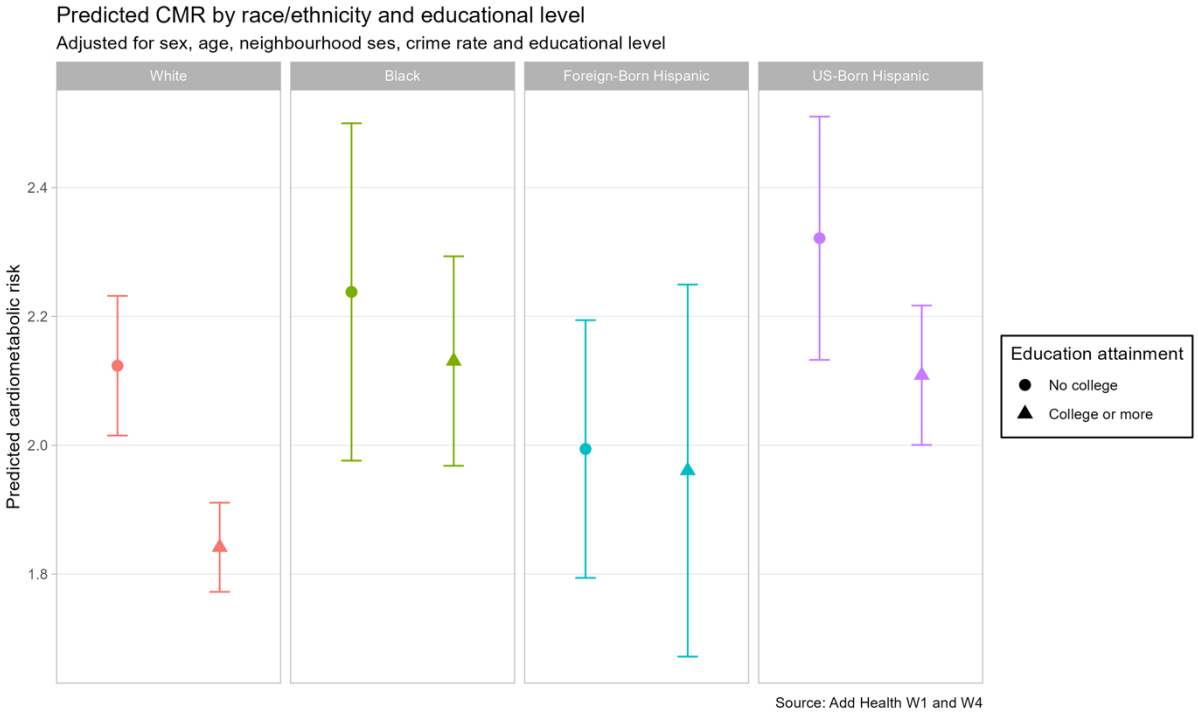

Figure 1 Predicted CMR by race/ethnicity, nativity and educational level

Figure 1 demonstrates that individuals with a college degree exhibit lower cardiometabolic risk (CMR) scores compared to those with less education, highlighting the general protective effect of higher educational attainment on health. This pattern is evident for White, Foreign-Born Hispanic, and US-Born Hispanic respondents, but appears weaker for Black respondents, supporting our first hypothesis.

White respondents show the clearest education-health gradient, with college-educated individuals having significantly lower predicted CMR scores than their less-educated counterparts. For Foreign-Born Hispanic respondents, while higher education is associated with lower CMR, the reduction is smaller and not statistically significant as indicated by the overlapping confidence intervals.

US-Born Hispanics display a substantial educational gradient, with college-educated individuals showing lower CMR scores compared to those without college education. In contrast, Black respondents show the smallest reduction in CMR associated with higher education, with the difference not reaching statistical significance.

These findings indicate that while education generally improves health outcomes across groups, its benefits are not uniform. The relatively flat education-health gradient for Black respondents is consistent with the “diminishing returns” hypothesis, suggesting that structural

and contextual factors may limit the extent to which educational attainment translates into health advantages for Black Americans.

Table 2. Poisson regression results of baseline and models with interactions

Variables	Baseline Model	Family Cohesion	Parent's Expectations on Study	Parenting Style	Parent Education	Parental Income	Two Parent Family
Constant	0.603***	0.590***	0.610***	0.607***	0.658***	0.621***	0.588***
	(0.066)	(0.070)	(0.063)	(0.065)	(0.066)	(0.065)	(0.069)
College+	-1.671***	-0.136***	-0.172***	-0.181***	-0.114***	-0.135***	-0.157***
	(0.023)	(0.044)	(0.026)	(0.034)	(0.032)	(0.032)	(0.035)
NH Black	0.034	0.050	-0.009	0.019	0.027	0.028	0.007
	(0.057)	(0.079)	(0.070)	(0.066)	(0.069)	(0.057)	(0.065)
NH Black ×College+	0.132**	0.098	0.181***	0.192***	0.088	0.077*	0.129
	(0.047)	(0.095)	(0.055)	(0.058)	(0.055)	(0.046)	(0.078)
NH Black ×College+ ×Family		-0.004	0.083	0.027	0.015	0.053	0.121*
		(0.063)	(0.052)	(0.061)	(0.102)	(0.169)	(0.071)
US-Born Hispanic	-0.066	-0.034	-0.158**	0.006	-0.042	-0.068	-0.021
	(0.058)	(0.156)	(0.062)	(0.092)	(0.053)	(0.057)	(0.091)
US-Born Hispanic ×College+	0.069	0.144	0.074	0.007	0.008	0.033	0.100
	(0.120)	(0.169)	(0.086)	(0.150)	(0.094)	(0.110)	(0.229)
US-Born Hispanic ×College+ ×Family		-0.022	0.326**	-0.224	-0.237	-0.219*	-0.083
		(0.174)	(0.139)	(0.280)	(0.269)	(0.119)	(0.120)
Foreign-Born Hispanic	0.120**	0.199***	0.099	0.172***	0.129***	0.111**	0.161**
	(0.045)	(0.054)	(0.060)	(0.065)	(0.048)	(0.048)	(0.068)
Foreign-Born Hispanic ×College+	0.036	-0.041	0.069	-0.043	-0.019	0.014	0.019
	(0.060)	(0.097)	(0.054)	(0.093)	(0.054)	(0.064)	(0.118)
Foreign-Born Hispanic×College+×Family		-0.111	0.053	-0.131	-0.154	0.040	-0.072
		(0.091)	(0.122)	(0.108)	(0.172)	(0.072)	(0.099)

*p<.05; **p<.01; ***p<.001

All models adjusted for personal and neighborhood characteristics.

4.3 The Contexts which Racial/Ethnic-Nativity Groups Experience the Highest Benefits of Educational Attainment

Table 2 shows the inclusion of interaction terms between educational attainment, race/ethnicity-nativity, and family environment factors. In the baseline model without family environment interactions, Black respondents with higher education show significantly smaller reductions in cardiometabolic risk compared to their White counterparts, as indicated by the positive Black×College+ coefficient. This finding aligns with the "diminishing returns" hypothesis, suggesting that Black Americans experience fewer health benefits from educational attainment than White Americans.

When family environment interactions are introduced, several important patterns emerge. The reduced significance of the Black×College+ interaction after accounting for family cohesion, parental education, parental income, and two-parent family structure suggests that these family factors partially explain the differential health returns to education between Black and White respondents. This indicates that family context plays an important role in moderating the education-health relationship for Black Americans.

For Foreign-Born Hispanics, the baseline model shows no significant difference in the education-health relationship compared to White respondents. However, when examining specific family contexts, a notable pattern emerges with Parents' Expectations on Study. Foreign-Born Hispanic respondents with higher education from families with high educational expectations show particularly weaker health benefits, and respondents whose parents have higher income show stronger health benefits.

US-Born Hispanics present a different pattern. While they show higher overall cardiometabolic risk compared to White respondents across most models, the education-health gradient does not differ significantly between these groups in the baseline model. The addition of family environment interactions does not substantially alter this similarity in educational health benefits, suggesting that US-Born Hispanics experience comparable health returns to education as White respondents, despite their higher baseline risk levels.

These interaction patterns highlight how the relationship between education and health operates differently across racial/ethnic-nativity groups and is contingent on various dimensions of family environment, partially supporting our second hypothesis.

Table 3. Poisson regression results of models stratified by family environments and race/ethnicity and nativity

Family Environment	Family Cohesion		Parents' Expectations on Study		Parenting Style		Parents' Education		Parental Income		Two Parents Family	
	High	Low	High	Low	Autonomy	Strict	College +	Less than College	High	Low	Yes	No
NH White (*n* = 5991)												
College or more	-0.158***	-0.128***	-0.143***	-0.148***	-0.124***	-0.170***	-0.181***	-0.097***	-0.185***	-0.116***	-0.155***	-0.139***
	(0.029)	(0.045)	(0.051)	(0.028)	(0.044)	(0.036)	(0.044)	(0.034)	(0.055)	(0.035)	(0.037)	(0.034)
Constant	0.364	0.790	0.760	0.497	0.571	0.516	0.317	0.731	0.555	0.564	0.381	0.802
	(0.107)	(0.129)	(0.172)	(0.079)	(0.118)	(0.111)	(0.148)	(0.105)	(0.158)	(0.077)	(0.108)	(0.129)
n	3,516	2,475	1,153	4,838	3,092	2,899	2,338	3,653	1,881	4,110	3,725	2,266
NH Black (*n* = 1,911)												
College or more	-0.084*	-0.079	-0.107*	-0.067	-0.145**	-0.026	-0.066	-0.094*	-0.035	-0.101**	-0.132*	-0.065
	(0.045)	(0.105)	(0.064)	(0.059)	(0.065)	(0.058)	(0.104)	(0.050)	(0.150)	(0.047)	(0.074)	(0.062)
Constant	0.680	0.995	0.977	0.693	0.954	0.662	0.596	0.902	0.646	0.854	0.706	0.848
	(0.187)	(0.291)	(0.191)	(0.193)	(0.224)	(0.184)	(0.284)	(0.165)	(0.348)	(0.175)	(0.281)	(0.165)
n	1,221	690	809	1,102	875	1,036	719	1,192	343	1,568	686	1,125
US-Born Hispanic (*n* = 982)												
College or more	-0.130*	-0.184*	-0.284**	-0.116*	-0.011	-0.246***	-0.033	-0.140***	-0.284***	-0.135*	-0.134*	-0.182
	(0.071)	(0.109)	(0.117)	(0.061)	(0.050)	(0.086)	(0.203)	(0.052)	(0.089)	(0.073)	(0.070)	(0.113)
Constant	1.109	1.455	1.791	1.096	1.518	1.047	1.254	1.252	1.077	1.312	0.853	1.898
	(0.306)	(0.310)	(0.486)	(0.283)	(0.273)	(0.306)	(0.417)	(0.175)	(0.570)	(0.195)	(0.225)	(0.311)
n	556	426	244	738	448	534	171	811	141	841	570	412
Foreign-Born Hispanic (*n* = 245)												
College or more	-0.125	0.035	-0.043	-0.097	-0.003	-0.145	-0.372**	-0.122	-0.863	-0.067	-0.094	-0.212
	(0.146)	(0.140)	(0.133)	(0.070)	(0.185)	(0.136)	(0.158)	(0.107)	(0.357)	(0.108)	(0.084)	(0.153)
Constant	-0.630	0.217	-3.936	0.448	1.076	-0.735	-0.938	-0.228	-0.039	-0.173	0.351	-1.323
	(0.707)	(1.018)	(1.010)	(0.616)	(0.553)	(0.657)	(1.730)	(0.645)	(1.403)	(0.425)	(0.837)	(1.153)
n	168	77	59	186	115	130	51	194	10	235	138	107

†*p*<.10; **p*<.05; ***p*<.01; ****p*<.001

All models adjusted for personal and neighborhood characteristics.

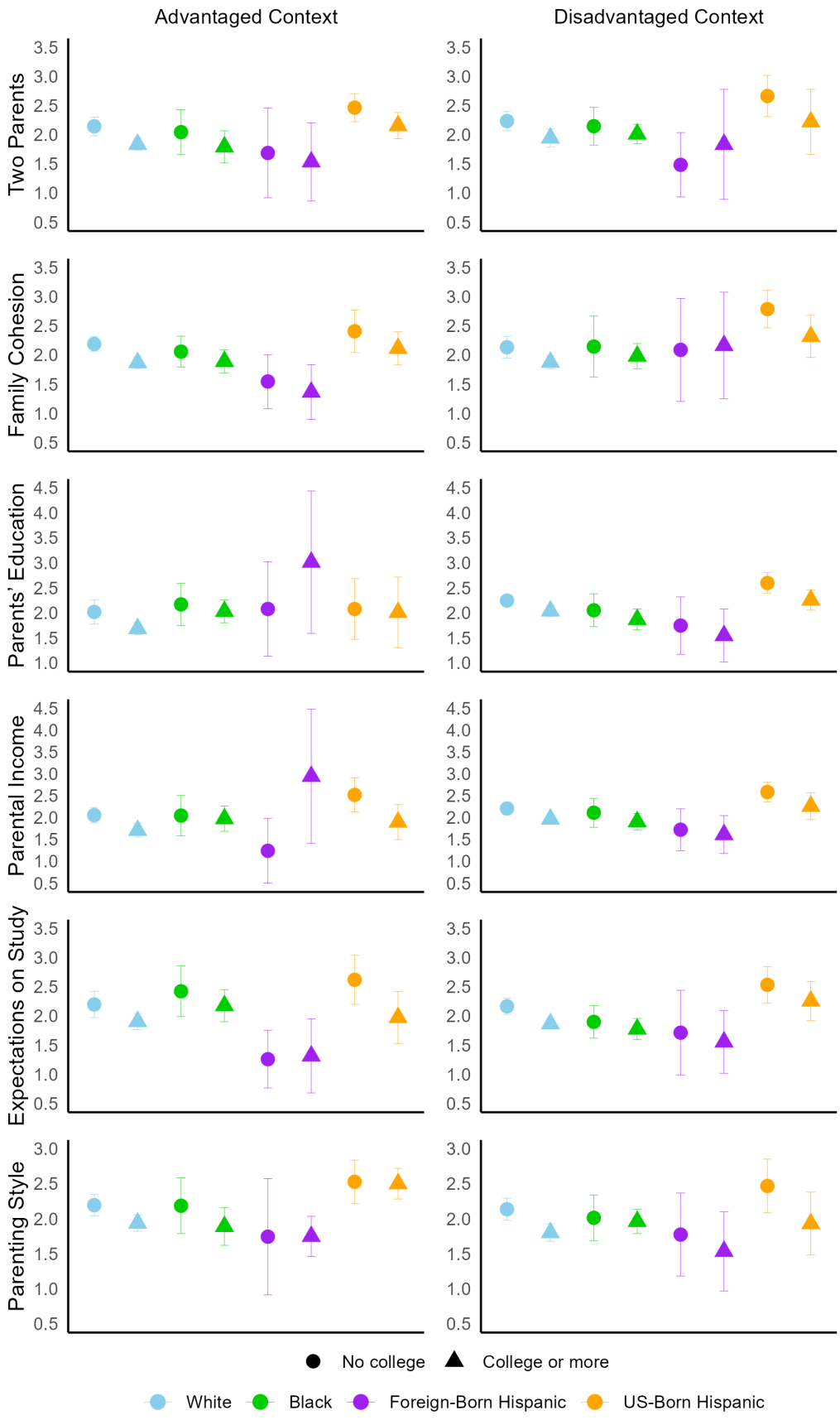

Figure 2 Predicted CMR Stratified by Race/Ethnicity and Family Environments

Table 3 presents findings from stratified analyses examining the association between educational attainment and CMR across racial/ethnic-nativity groups and family environment contexts. Figure 2 further illustrates these patterns.

Among White respondents, higher education is consistently associated with lower CMR across all family environments. The education-health relationship remains significant regardless of whether family environments are advantaged or disadvantaged. The strongest protective effects of education are observed in families with high parental income, strict parenting styles, and high family cohesion. Even in less advantaged contexts—such as families with lower parental education or single-parent structures—education provides significant cardiometabolic benefits. This suggests that for White Americans, the health advantages of higher education transcend early life family circumstances.

For Black respondents, the education-health relationship is more context-dependent. Higher education is significantly associated with lower CMR in families characterized by high cohesion, high parental expectations for education, and autonomy parenting styles. Educational attainment is also more protective for health among Black respondents from families with less educated parents, lower parental income, and two-parent structures. The education-health relationship is not significant in contexts of low family cohesion, low educational expectations, strict parenting, college-educated parents, or high parental income. These patterns suggest that for Black Americans, the health benefits of higher education are concentrated among those from specific family backgrounds rather than being universally protective.

US-Born Hispanic respondents show significant education-health associations across most family contexts, with varying magnitudes. The education-health relationship is particularly strong in families with high parental income, high educational expectations, and strict parenting styles. In contrast, educational attainment shows no significant health benefits for US-Born Hispanics from families with autonomy parenting styles or college-educated parents. This suggests that for US-Born Hispanics, the translation of educational attainment into health benefits is enhanced in certain advantaged family contexts but constrained in others.

Foreign-Born Hispanic respondents display the most limited pattern of educational benefits. Higher education is significantly associated with lower CMR only among those with college-educated parents. The education-health relationship is not significant across other family environments. This restricted pattern could reflect the smaller sample size for this group, but may also indicate that for immigrant populations, the relationship between education and health is shaped by unique processes related to migration experiences and acculturation.

Figure 2 illustrates these patterns visually, showing how predicted CMR varies by education level across different family contexts for each racial/ethnic-nativity group. The figure demonstrates that the education-health relationship is consistently strong for White respondents regardless of family context, while showing greater variability for Black, US-Born Hispanic, and Foreign-Born Hispanic respondents.

Discussion

This study examined how early life family environments moderate the relationship between educational attainment and cardiometabolic risk across racial/ethnic-nativity groups. Our findings reveal complex patterns that challenge universal understandings of how education relates to health and underscore the importance of considering family context when investigating health disparities.

Our baseline analysis confirms that individuals with higher educational attainment generally experience lower cardiometabolic risk, though this relationship varies significantly by race/ethnicity. White respondents show the strongest educational gradient in health, followed by US-Born Hispanics, while Black respondents demonstrate the weakest association between education and health. This pattern aligns with prior research documenting "diminishing returns" of education for Black Americans (Assari et al., 2017; Farmer et al., 2021).

Our findings suggest that family environment significantly moderates these educational disparities in health in ways that differ markedly across racial/ethnic-nativity groups. For Black Americans, our findings reveal both intuitive and counterintuitive patterns. As expected, higher education was associated with better health for Black respondents from families with high cohesion, high educational expectations, and two-parent structures. These supportive environments may provide emotional resources that buffer stresses associated with educational attainment (Taylor & Seeman, 1999).

However, higher education was associated with better health for Black respondents from families with less educated parents, lower income, and "natural growth" parenting styles—but not for those from more socioeconomically advantaged backgrounds. Several mechanisms might explain these findings. First, Black individuals from disadvantaged backgrounds who achieve higher education may develop exceptional resilience and coping resources that benefit health (Brody et al., 2013). Second, the contrast between pre-college disadvantage and post-college advantage may produce greater relative health benefits. Third, Black Americans from advantaged backgrounds may face unique stressors navigating predominantly White educational and professional spaces, including "cultural taxation" and the psychological costs of code-switching (Geronimus et al., 2006; Hirshfield & Joseph, 2012).

For US-Born Hispanics, the health benefits of education were most pronounced in family environments characterized by high educational expectations, high income, and "concerted cultivation" parenting styles. These patterns suggest that family environments emphasizing and supporting educational achievement enhance health returns to higher education for this group. The absence of educational health benefits among those from families with college-educated parents may reflect complex dynamics related to assimilation processes or ceiling effects.

Foreign-Born Hispanics displayed a limited pattern of educational health benefits, with higher education significantly associated with lower CMR only among those with college-educated parents. This restricted pattern suggests that for immigrant populations, intergenerational

educational resources may be particularly important for translating educational attainment into health advantages (Portes & Fernández-Kelly, 2008). The "healthy immigrant effect" may also buffer health regardless of family environment or educational attainment (Markides & Rote, 2015).

For White Americans, higher education was consistently associated with better cardiometabolic health across all family environments, suggesting that educational health advantages transcend early life family circumstances for this group. This robust relationship is consistent with the Theory of Fundamental Social Causes, indicating that White Americans face fewer structural barriers in translating educational attainment into health advantages (Link & Phelan, 1995).

These findings have important theoretical implications. They challenge the notion of education as a "great equalizer" and extend the Theory of Fundamental Social Causes by demonstrating that the mechanisms linking education to health vary not only across time and place but also across family contexts within the same time and place. They also underscore the limitations of individual-level explanations for educational health disparities.

From a policy perspective, interventions aimed at reducing health disparities should consider the family contexts in which educational attainment occurs. Programs supporting diverse family structures, providing resources to disadvantaged families, and addressing structural barriers faced by minoritized groups may enhance the health returns to education across populations.

Several limitations should be considered when interpreting our findings. Our analysis cannot fully elucidate the mechanisms through which family environments moderate the education-health relationship. Our sample of Foreign-Born Hispanics was relatively small, potentially limiting our ability to detect significant patterns. Our focus on young adulthood does not reveal how these patterns might evolve over the life course. Despite these limitations, this study provides important insights into the complex interplay between family background, education, and health across diverse populations.

References

Assari, S. (2018). Health disparities due to diminished return among Black Americans: Public policy solutions. *Social Issues and Policy Review*, *12*(1), 112–145. https://doi.org/10.1111/sipr.12042

Assari, S., Nikahd, A., Malekahmadi, M. R., Lankarani, M. M., & Zamanian, H. (2016). Race by Gender Group Differences in the Protective Effects of Socioeconomic Factors Against Sustained Health Problems Across Five Domains. *Journal of Racial and Ethnic Health Disparities*. https://doi.org/10.1007/s40615-016-0291-3

Beltrán-Sánchez, H., Palloni, A., Riosmena, F., & Wong, R. (2016). SES Gradients Among Mexicans in the United States and in Mexico: A New Twist to the Hispanic Paradox? *Demography*, *53*(5), 1555–1581. https://doi.org/10.1007/s13524-016-0508-4

Boen, C. (2016). The Role of Socioeconomic Factors in Black-White Health Inequities Across the Life Course: Point-in-time Measures, Long-term Exposures, and Differential Health Returns. *Social Science & Medicine (1982)*, *170*, 63–76. https://doi.org/10.1016/j.socscimed.2016.10.008

Brody, G. H., Yu, T., Chen, E., Miller, G. E., Kogan, S. M., & Beach, S. R. H. (2013). Is resilience only skin deep? Rural African Americans' socioeconomic status-related risk and competence in preadolescence and psychological adjustment and allostatic load at age 19. *Psychological Science*, *24*, 1285–1293.

Chetty, R., Hendren, N., Jones, M. R., & Porter, S. R. (2020). Race and Economic Opportunity in the United States: An Intergenerational Perspective*. *The Quarterly Journal of Economics*, *135*(2), 711–783. https://doi.org/10.1093/qje/qjz042

Colen, C. G., Ramey, D. M., Cooksey, E. C., & Williams, D. R. (2018). Racial disparities in health among nonpoor African Americans and Hispanics: The role of acute and chronic discrimination. *Social Science & Medicine (1982)*, *199*, 167–180. https://doi.org/10.1016/j.socscimed.2017.04.051

Conti, G., Heckman, J., & Urzua, S. (2010). The Education-Health Gradient. *American Economic Review*, *100*(2), 234–238. https://doi.org/10.1257/aer.100.2.234

Darling, N., & Steinberg, L. (1993). Parenting style as context: An integrative model. *Psychological Bulletin*, *113*(3), 487–496. https://doi.org/10.1037/0033-2909.113.3.487

Farmer, H. R., Wray, L. A., & Haas, S. A. (2020). Race, Gender, and Socioeconomic Variations in C-Reactive Protein Using the Health and Retirement Study. *The Journals of Gerontology Series B: Psychological Sciences and Social Sciences*, *76*(3), 583–595. https://doi.org/10.1093/geronb/gbaa027

Friedman, E. M., Montez, J. K., Sheehan, C. M., Guenewald, T. L., & Seeman, T. E. (2015). Childhood Adversities and Adult Cardiometabolic Health: Does the Quantity, Timing, and Type of Adversity Matter? *Journal of Aging and Health*, *27*(8), 1311–1338. https://doi.org/10.1177/0898264315580122

Fuller-Rowell, T. E., Curtis, D. S., Doan, S. N., & Coe, C. L. (2015). Racial disparities in the health benefits of educational attainment: A study of inflammatory trajectories among African American and White adults. *Psychosomatic Medicine*, *77*, 33–40.

Gaydosh, L., & Harris, K. M. (2024). Institutional Context Shapes the Physical Health of College Graduates Differently for U.S. White, Black, and Hispanic Adults. *Demography*, *61*(3), 933–966. https://doi.org/10.1215/00703370-11380743

Geronimus, A. T., James, S. A., Destin, M., Graham, L. F., Hatzenbuehler, M. L., Murphy, M. C., & Thompson, J. P. (2016). Jedi public health: Co-creating an identity-safe culture to promote health equity. *SSM-Population Health*, *2*, 105–116.

Goldman, N., Kimbro, R. T., Turra, C. M., & Pebley, A. R. (2006). Socioeconomic gradients in health for White and Mexican-origin populations. *American Journal of Public Health*, *96*, 2186–2193.

Hardie, J. H., & Seltzer, J. A. (2016). Parent-Child Relationships at the Transition to Adulthood: A Comparison of Black, Hispanic, and White Immigrant and Native-Born Youth. *Social Forces; a Scientific Medium of Social Study and Interpretation*, *95*(1), 321–354. https://doi.org/10.1093/sf/sow033

Hargrove, T. W. (2018). Intersecting social inequalities and body mass index trajectories from adolescence to early adulthood. *Journal of Health and Social Behavior*, *59*, 56–73.

Hargrove, T. W., Gaydosh, L., & Dennis, A. C. (2022). Contextualizing Educational Disparities in Health: Variations by Race/Ethnicity, Nativity, and County-Level Characteristics. *Demography*, *59*(1), 267–292. https://doi.org/10.1215/00703370-9664206

Harris, K. M. (2010). An integrative approach to health. *Demography*, *47*, 1–22.

Harris, K. M., & McDade, T. W. (2018). The biosocial approach to human development, behavior, and health across the life course. *Russell Sage Foundation Journal of the Social Sciences*, *4*, 2–26.

Hayward, M. D., Hummer, R. A., & Sasson, I. (2015). Trends and group differences in the association between educational attainment and U.S. adult mortality: Implications for understanding education's causal influence. *Social Science & Medicine*, *127*, 8–18.

Hudson, D. L., Neighbors, H. W., Geronimus, A. T., & Jackson, J. S. (2012). Are benefits conferred with greater socioeconomic position undermined by racial discrimination among African American men? *Journal of Men's Health*, *9*, 127–136.

Hudson, D. L., Neighbors, H. W., Geronimus, A. T., & Jackson, J. S. (2016). Racial discrimination, John Henryism, and depression among African Americans. *Journal of Black Psychology*, *42*, 221–243.

James, S. A., Hartnett, S. A., & Kalsbeek, W. D. (1983). John Henryism and blood pressure differences among Black men. *Journal of Behavioral Medicine*, *6*, 259–278.

Kennedy, S., Kidd, M. P., McDonald, J. T., & Biddle, N. (2015). The healthy immigrant effect: Patterns and evidence from four countries. *Journal of International Migration and Integration*, *16*, 317–332.

Lareau, A. (2011). *Unequal Childhoods: Class, Race, and Family Life*. University of California Press.

Link, B. G., & Phelan, J. (1995). Social conditions as fundamental causes of disease. *Journal of Health and Social Behavior*, *Extra Issue*, 80–94.

Markides, K. S., & Rote, S. (2015). Immigrant Health Paradox. In R. A. Scott & S. M. Kosslyn (Eds.), *Emerging Trends in the Social and Behavioral Sciences* (1st ed., pp. 1–15). Wiley. https://doi.org/10.1002/9781118900772.etrds0174

McLanahan, S., & Percheski, C. (2008). Family structure and the reproduction of inequalities. *Annual Review of Sociology*, *34*, 257–276. https://doi.org/10.1146/annurev.soc.34.040507.134549

Miller, G. E., Cohen, S., Janicki-Deverts, D., Brody, G. H., & Chen, E. (2016). Viral challenge reveals further evidence of skin-deep resilience in African Americans from disadvantaged backgrounds. *Health Psychology*, *35*, 1225–1234.

Montez, J. K., & Bisesti, E. M. (2024). Widening Educational Disparities in Health and Longevity. *Annual Review of Sociology*, *50*(1), 547–564. https://doi.org/10.1146/annurev-soc-071723-080605

Montez, J. K., & Friedman, E. M. (2015). Educational attainment and adult health: Under what conditions is the association causal? *Social Science & Medicine*, *127*, 1–7.

Montez, J. K., & Hayward, M. D. (2014). Cumulative childhood adversity, educational attainment, and active life expectancy among U.S. adults. *Demography*, *51*(2), 413–435. https://doi.org/10.1007/s13524-013-0261-x

Montez, J. K., Hayward, M. D., & Zajacova, A. (2019). Educational Disparities in Adult Health: U.S. States as Institutional Actors on the Association. *Socius*, *5*, 2378023119835345. https://doi.org/10.1177/2378023119835345

Montez, J. K., Hummer, R. A., & Hayward, M. D. (2012). Educational Attainment and Adult Mortality in the United States: A Systematic Analysis of Functional Form. *Demography*, *49*(1), 315–336. https://doi.org/10.1007/s13524-011-0082-8

Montez, J. K., Zajacova, A., Hayward, M. D., Woolf, S. H., Chapman, D., & Beckfield, J. (2019). Educational Disparities in Adult Mortality Across U.S. States: How Do They Differ, and Have They Changed Since the Mid-1980s? *Demography*, *56*(2), 621–644. https://doi.org/10.1007/s13524-018-0750-z

Phelan, J. C., & Link, B. G. (2015). Is Racism a Fundamental Cause of Inequalities in Health? *Annual Review of Sociology*, *41*(Volume 41, 2015), 311–330. https://doi.org/10.1146/annurev-soc-073014-112305

Riosmena, F., Wong, R., & Palloni, A. (2013). Migration selection, protection, and acculturation in health: A binational perspective on older adults. *Demography*, *50*, 1039–1064.

Rothon, C., Goodwin, L., & Stansfeld, S. (2012). Family social support, community "social capital" and adolescents' mental health and educational outcomes: A longitudinal study in England. *Social Psychiatry and Psychiatric Epidemiology*, *47*(5), 697–709. https://doi.org/10.1007/s00127-011-0391-7

Williams, D. R., Priest, N., & Anderson, N. B. (2016). Understanding associations among race, socioeconomic status, and health: Patterns and prospects. *Health Psychology*, *35*, 407–411.

Zajacova, A., & Lawrence, E. M. (2018). The relationship between education and health: Reducing disparities through a contextual approach. *Annual Review of Public Health*, *39*, 273–289. https://doi.org/10.1146/annurev-publhealth-031816-044628